

Unit 1, Lesson 1: Using Multiple Representations to Investigate Patterns

Benchmarks

MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.

MA.5.AR.3.2 Given a rule for a numerical pattern, use a two-column table to record the inputs and outputs.

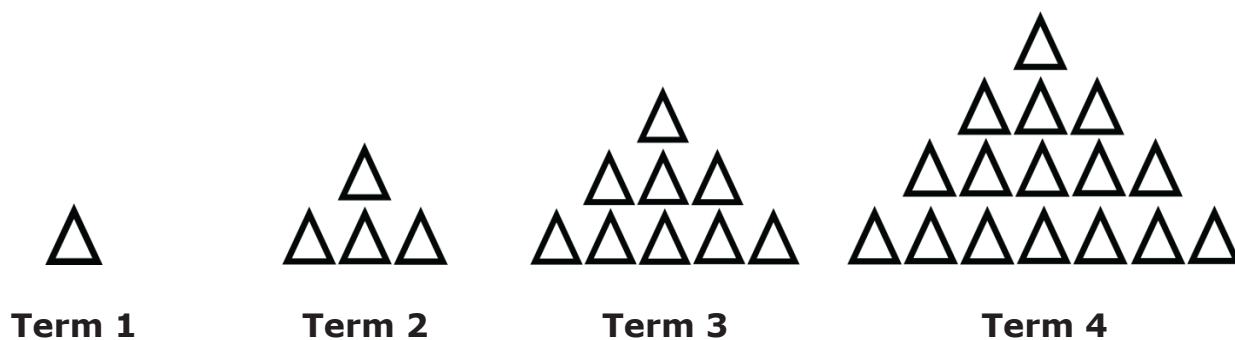
MA.5.GR.4.2 Represent mathematical and real-world problems by plotting points in the first quadrant of the coordinate plane and interpret coordinate values of points in the context of the situation.

Learning Targets

- ☐ I can use multiple representations to investigate patterns found in tables or lists.
- ☐ I can plot points on the coordinate plane.

1.1.1 Warm-Up: Visual Patterns

The diagram shows four terms of a visual pattern.



1. What do you notice about the pattern?
2. How many rows of triangles do you think will be in Term 9?
3. How many of the small triangles do you think will be in the pattern for Term 9?
4. How many of the small triangles do you think will be in the pattern for Term 15?

1.1.2 Guided Instruction: Using Multiple Representations to Investigate Patterns

1. A team of mathematicians at an orange juice manufacturer in Florida is collecting information on how many oranges are needed to make orange juice.

The table shows the information the mathematicians have recorded so far.

Cartons of Orange Juice	Average Number of Oranges Needed

- a. What do you notice about the relationship between the cartons of orange juice and the number of oranges needed?
- b. What is the average number of oranges needed for cartons of orange juice?
- c. A healthy, mature orange tree produces between and oranges. What is the range of the number of cartons of orange juice that can be made using oranges from a healthy, mature orange tree?
- d. Write an equation that represents the relationship between the number of cartons of orange juice, , and , the average number of oranges needed.
- e. Use your equation or the pattern to complete the table.

Cartons of Orange Juice	Average Number of Oranges Needed

Each pair of values in the table can be plotted on the coordinate plane. The coordinate plane has two axes. The **-axis** is the horizontal axis and the **-axis** is the vertical axis.

Each pair of values becomes an **ordered pair**, where the -value is the first number in the ordered pair, and the -value is the second number in the ordered pair.

For example, the mathematicians found that making cartons of orange juice usually uses oranges.

f. Complete the statement.

The ordered pair that represents this relationship is

- (2, 40).
 - (40, 2).

To plot the -value, begin at the **origin**,

, then move

- up 2 units.
 - right 2 units.
 - right 40 units.
 - up 40 units.

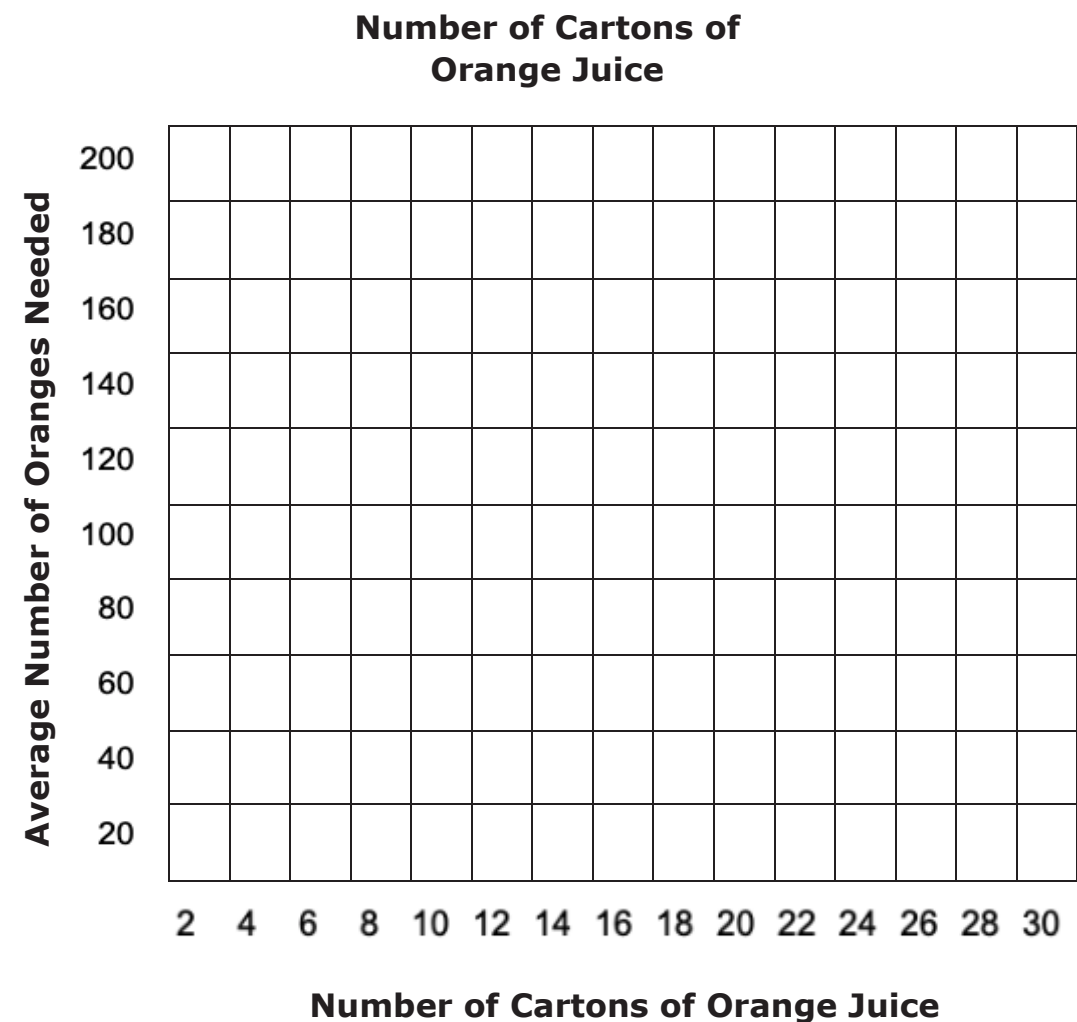
To plot the -value,

move

- up 2 units
 - right 2 units
 - right 40 units
 - up 40 units

from the -value.

- g. After discussing with your partner, write a description of how to plot the point for 1 carton of orange juice.
- h. Plot each point from the table on the coordinate plane below.



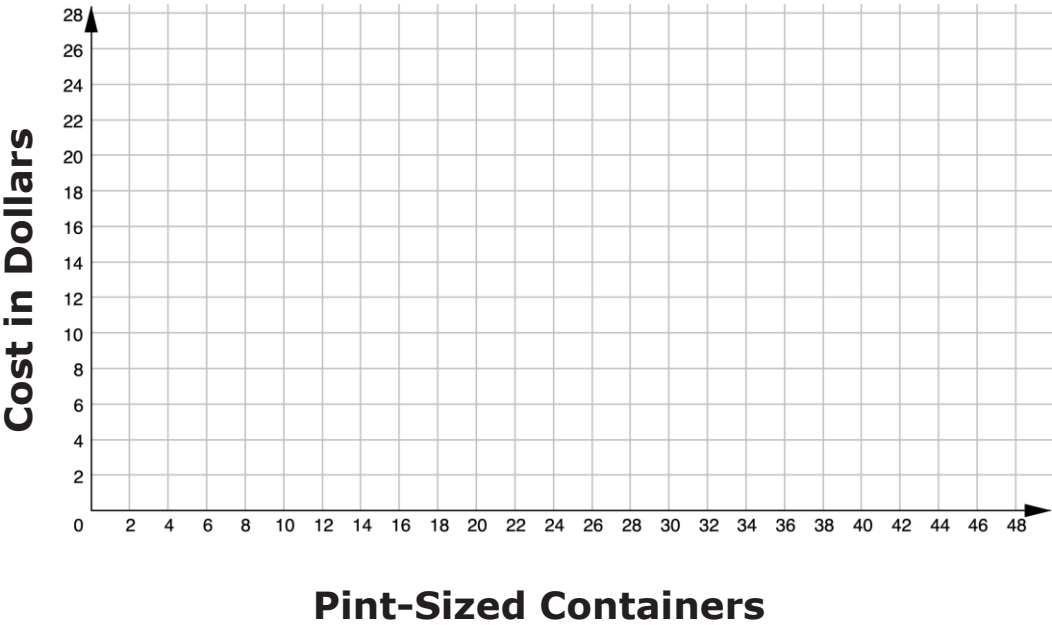
- i. Use the table, equation, or graph to complete the ordered pair.
- j. What does each number in the ordered pair represent?

1.1.3 Guided Practice: Investigating Patterns

1. The orange juice manufacturer decided to offer orange juice in pint-sized containers. Each -pack of pint-sized containers will be sold for .
- a. Use the pattern in the table to find the missing values.

Pint-Sized Containers	Cost (\$)

- b. Write an equation to represent the relationship between the number of pint-sized containers, , and the cost in dollars, .
- c. Use the coordinate grid to plot the ordered pairs in the table.



d. Use the table, the equation, or the graph to determine the cost of pint-sized containers.

e. Which representation did you use?

2. Elena, Kimiko, and Vicente were discussing the two graphs.

Vicente: In the first graph, would the ordered pair fit the pattern?

Elena: I don't think so. Having cartons does not make sense.

Kimiko: The ordered pair for the second graph does make sense. That would be pint-size containers costing .

Elena: No, Kimiko, the is , and the is pint-size containers. A pint-size container would not make sense.

a. Explain whether or not you think the ordered pairs and make sense for the contexts they represent.

b. Ask your partner to explain their answer to you. Write down their explanation.

1.1.4 Wrap-Up: Reflection

1. For each pair of statements, check the statements that apply to your understanding.
 - ☐ I can use an expression to extend patterns.
 - ☐ I need support using an expression to extend patterns.
 - ☐ I can locate ordered pairs on a coordinate plane.
 - ☐ I need support in locating ordered pairs on a coordinate plane.
 - ☐ I can identify a pattern and write an expression to match.
 - ☐ I need support identifying a pattern and writing an expression to match.